

Which Chinese Standard? Mainland and Taiwan Grade 65% of Shared Vocabulary Differently, and No Alignment Fixes It

Alvaro Serrano

Independent researcher

alvaro.serrano.gp@gmail.com

orcid.org/0009-0006-4701-9026

Abstract

Chinese has two national proficiency ecosystems. The mainland publishes a grading standard and an examination syllabus; Taiwan publishes TBCL and TOCFL. Learners, publishers and researchers move between them constantly, but the published comparisons are exam-choice guidance — level equivalency tables and headline vocabulary counts — and no word-by-word comparison of the inventories appears to exist. We provide one. Of the 8,318 words TBCL and the 2025 mainland syllabus share, **64.7% sit at different tiers**; across the character inventories the figure is 55.2%. Both far exceed the 41.5% disagreement between the mainland’s *own* two documents (Serrano, 2026), which our pipeline reproduces exactly as a control. We also report a claim we cannot make. The obvious next question — which side grades harder — appears to have an emphatic answer, 4,783 words against 598. It does not survive: TBCL levels 1 and 2 sit below CEFR A1 while the mainland scale has no sub-A1 tier, and shifting TBCL by a single level reverses the direction to 1,234 against 2,421. The disagreement rate holds under every alignment tested, between 43.9% and 96.2%; the direction is an artifact of a correspondence no issuing body has published. We release the extraction and comparison code, and a validated machine-readable extraction procedure for TBCL, for which none appears to have existed.

1 Two ecosystems, never compared

A learner who studies for HSK and then sits TOCFL, a publisher levelling a series for both markets, or a researcher assembling a Chinese resource, all face the same question: do the two standards mean the same thing by a level? The literature does not answer it. What ex-

	Document	Words	Chars
PRC	GF0025-2021 standard	10,916	3,000
	2025 exam syllabus	10,896	3,088
TW	TBCL standard	14,452	3,100
	TOCFL 8000 exam list	7,517	—

Table 1: The four inventories. Taiwan figures are as published; TBCL’s word list carried 14,425 entries before the April 2025 revision.

ists is comparison for the purpose of choosing an examination — equivalency tables asserting that HSK 4 is “about” TOCFL 3, and totals for how many words each requires. Those are statements about scales, not about vocabulary.

The question matters more than it did. The mainland’s 2025 examination syllabus took effect in July 2026, and its arrival prompted a widely repeated claim that mainland word counts have now overtaken Taiwan’s. That claim turns out to be true and false at once, depending on a distinction nobody making it draws (§2).

We compare the inventories directly, in the manner of Serrano (2026), which measured the same disagreement between the mainland’s two documents.

2 The four documents

Table 1 settles the overtaking claim, and shows why it was ambiguous. Compared exam to exam, the mainland syllabus is 45% larger than TOCFL’s list and the claim holds. Compared standard to standard, TBCL is 32% larger than the mainland standard and the claim fails. The two comparisons give opposite answers, and the sources making the claim do not say which pair they mean — the same failure of document identity that Serrano (2026) documents within the mainland, now on the

cross-strait axis.

Extracting TBCL. No machine-readable extraction of TBCL appears to exist. NAER publishes the lists as spreadsheets and asserts rights over them, so we release the extraction procedure and not the data. One detail defeats the obvious approach. Levels 1 to 4 each appear in two forms, 第 N 級 and 第 N* 級, while levels 5 to 7 do not; filtering on the unstarred form — which is what a reader who has not noticed the asterisk will write — silently discards 442 of 3,100 characters and 1,398 of 14,452 words, and produces totals that look plausible. Counting both forms makes level 1 exactly 246 characters, the figure NAER publishes, which is how we established that starred entries belong to their numbered level. **What the asterisk denotes we do not determine.** It does not track the word list’s own 核心詞 core-word marking: both forms contain core and non-core entries. Our extraction records the flag and leaves the question open.

3 Method

Each Taiwan inventory is converted to simplified characters with OpenCC before matching. Conversion is many-to-one, and 23 traditional entries collapse onto a simplified form already present; we report this rather than absorb it. TBCL writes alternates with a slash — 爸爸/爸, 姊姊/姐姐/姊/姐 — in 365 entries, and we match each variant separately while counting the entry once.

Aligning the scales. TBCL has seven levels. The mainland standards have nine but grade 7, 8 and 9 as one undifferentiated band, so they have seven distinct tiers. We align tier 7 to tier 7 as “the top band” and levels 1–6 one to one. **This is an assumption, not a finding.** No issuing body publishes a TBCL-to-HSK correspondence, and §5 shows how much rests on it.

Control. Before comparing across the strait we ran the same pipeline over the two mainland documents, where the answer is published. It reproduces Serrano (2026) exactly: 9,698 shared words with 41.5% at different tiers, and 2,945 shared characters with 40.7%. The cross-strait figures below come from the same code path.

Comparison	Shared	Differ	Rate
<i>Words</i>			
PRC 2021 vs 2025	9,698	4,023	41.5%
TBCL vs PRC 2025	8,318	5,381	64.7%
TBCL vs PRC 2021	7,885	5,607	71.1%
<i>Characters</i>			
PRC 2021 vs 2025	2,945	1,200	40.7%
TBCL vs PRC 2025	2,785	1,538	55.2%
TBCL vs PRC 2021	2,737	1,601	58.5%

Table 2: Shared items graded at different tiers.

Alignment	Differ	PRC harder	TBCL harder
1:1 (as reported)	64.7%	4,783	598
TBCL −1 level	91.2%	7,497	85
TBCL −2 levels	96.2%	7,982	19
TBCL +1 level	43.9%	1,234	2,421

Table 3: Shifting one scale against the other. The rate never falls below the mainland’s internal 41.5%; the direction reverses.

4 Results

The two ecosystems disagree with each other substantially more than the mainland’s own two documents do: 64.7% of shared vocabulary against 41.5%, and 55.2% of shared characters against 40.7%. The gap is not explained by inventory size, since TBCL and the 2025 syllabus share 8,318 words — a comparable base to the mainland pair’s 9,674.

A practical reading: a text levelled under one ecosystem carries almost no information about its level under the other. Two thirds of the vocabulary that determines the answer is graded differently.

5 The direction does not survive

Under the 1:1 alignment the mainland syllabus appears to grade shared vocabulary harder by 4,783 words to 598 — a margin that invites a conclusion about pedagogical conservatism. We do not draw it.

TBCL levels 1 and 2 sit below CEFR A1 in listening, reading and writing, while the mainland scale describes no sub-A1 tier. If the scales are offset by a level, a 1:1 alignment manufactures a direction. Table 3 shows that shifting TBCL by one level in the other direction reverses it entirely, to 1,234 against 2,421.

What survives is the rate. Under every alignment tested it stays between 43.9% and

96.2%, and its minimum still exceeds the 41.5% disagreement inside the mainland pair. **The finding is that the two ecosystems disagree; which one is harder is not determinable without a correspondence that does not exist.**

That absence is itself the result worth carrying. TOCFL score reports have carried a TBCL column since August 2021, so Taiwan publishes an exam-to-standard correspondence internally. Nothing comparable exists across the strait, and the equivalency tables in circulation are reconstructions.

6 Limitations

The level alignment is the largest, and §5 is the response to it rather than a defence. Simplification is many-to-one and 23 entries collapse; a comparison in the traditional direction would lose more, since the mainland lists are published simplified. Matching is by orthographic form, so a word shared in writing but differing in sense or register counts as shared — the figures are therefore a lower bound on disagreement, not an upper one. TOCFL is not compared item by item here; we use it only for the count in §2. TBCL’s starred levels are counted at their numbered level on the evidence of one published total, and a different reading of the asterisk would change the per-level figures though not the totals. Finally, both Taiwan inventories were extracted by us and validated against published totals, but not against an official machine-readable release, because none exists.

Availability

`scripts/tbcl_extract.py` and `scripts/crossstrait.py` in the `hsk30` repository reproduce every figure here from spreadsheets downloaded free from the issuing bodies. Neither Taiwan inventory is redistributed: NAER and SC-TOP assert rights over them, so the code ships and the data does not. The mainland lists, the grading library and the corpus are released as described in [Serrano \(2026\)](#).

Ethics and data statement

No personal data and no human subjects. All four inventories are published national refer-

ence documents, used here to compare those documents with each other. Generative AI was used in writing the extraction and comparison code and in drafting this paper; the problem framing, the design decisions and every claim are the author’s, who reviewed all output. That review is what identified the alignment artifact reported in §5.

References

Alvaro Serrano. 2026. [Which HSK 3.0? two official documents, and half of all grading decisions change](#). Preprint.